

## STEPCLIMBER · REQUIREMENTS

# StepClimber requirements

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## DOCUMENT CONTROL

|             |                                            |         |                |
|-------------|--------------------------------------------|---------|----------------|
| Document ID | SCM-REQ-001                                | Version | 0.2            |
| Status      | <b>Draft</b>                               | Date    | 2026-09-25     |
| Author      | Amish Chadha                               | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 2</b> Technology concept formulated |         |                |
| Repository  | github.com/BoujeeEnjinia1701/stepclimber   |         |                |
| Source      | docs/03-requirements.md @ d20b7b7          |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                             |
|---------|------------|--------------|------------------------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                   |

## StepClimber requirements

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These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with couriers, and will be checked by calculation at TRL 3 and revised after user research (see SCM-PRB-001). The status column compares each target with the first-order estimates in SCM-PRC-001; no requirement has been verified.

The **design load case** is a 60 kg (132 lb) payload on a truck of about 24 kg, about 84 kg in total, climbing a residential stair with 196 mm (7.75 in) risers and 254 mm (10 in) treads, the steepest the US residential code allows.

| ID  | Requirement           | Target                                                                                                                                                                                                                                                                                                   | Status at TRL 2                                                                                                                 | Verification (TRL 3 or later)                 |
|-----|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| R1  | Rated stair load      | 60 kg (132 lb) payload up and down stairs; 100 kg on the flat                                                                                                                                                                                                                                            | By design; structure unverified                                                                                                 | Load and torque calculation; later proof load |
| R2  | Stair range           | Straight flights with risers 100 to 200 mm, treads 250 mm or more and nosing overhang up to 32 mm                                                                                                                                                                                                        | <b>At risk.</b> Fixed tri-star geometry; stairs steeper than this (for example 241 mm industrial risers) are <b>not</b> covered | Cluster geometry study against a stair survey |
| R3  | Climb speed           | 20 steps/min or more, up and down, at rated load                                                                                                                                                                                                                                                         | Met on power estimate (about 236 W peak motor output against 250 W rated); thin margin                                          | Drive calculation; later timed climb          |
| R4  | Endurance             | 1,000 or more loaded steps up plus 1,000 down per charge                                                                                                                                                                                                                                                 | Met on estimate (about 1,300)                                                                                                   | Energy calculation; later logged use          |
| R5  | Truck mass            | 25 kg or less with battery; battery removable and 3 kg or less                                                                                                                                                                                                                                           | Met on estimate (about 24.3 kg, pack about 2.6 kg); thin margin                                                                 | Mass roll-up; later weighing                  |
| R6  | Operator handle force | 100 N or less at the grip while climbing inside the tilt window (R7)                                                                                                                                                                                                                                     | Met on estimate (about 91 N at the window edge)                                                                                 | Static balance calculation; later force gauge |
| R7  | Tilt control          | IMU measures frame angle at 100 Hz or faster; the controller stores the balance angle when a climb starts, shapes motor speed to keep the angle within plus or minus 8 degrees of it, and stops and alerts within 0.2 s if the angle leaves that window or the range 15 to 45 degrees back from vertical | Concept only; unverified                                                                                                        | Control design review; later bench test       |
| R8  | Hold on any loss      | Load holds on the stair with power off, grip released, battery removed or any fault; drift 5 mm or less in 10 min at rated load                                                                                                                                                                          | By design (self-locking worm plus spring-applied brake); unverified                                                             | Brake and worm calculation; later hold test   |
| R9  | Hold-to-run           | Motion only while the dead-man grip is held and a direction is selected; release stops the clusters within 0.2 s                                                                                                                                                                                         | By design; unverified                                                                                                           | Circuit review; later bench test              |
| R10 | Flat rolling          | Push force 40 N or less at rated load on a smooth floor at walking pace, with the clusters parked                                                                                                                                                                                                        | Met on estimate (about 25 N)                                                                                                    | Rolling calculation; later pull test          |
| R11 | Size                  | Overall width 600 mm or less; upright height 1,300 mm or less to fit a car trunk or van shelf                                                                                                                                                                                                            | Width met (about 570 mm). <b>Height not met</b> (about 1,460 mm)                                                                | Model check                                   |
| R12 | Affordable            | Parts cost \$600 or less for one prototype, charger included                                                                                                                                                                                                                                             | Met on estimate (about \$580, indicative); about 3 % margin                                                                     | Priced BOM ( bom/ bom.csv )                   |

| ID  | Requirement                   | Target                                                                                                                | Status at TRL 2                                       | Verification (TRL 3 or later)      |
|-----|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------|
| R13 | Battery                       | 24 V class LiFePO4 with BMS and fused output; charging blocked below 0 °C and above 45 °C; full charge in 4 h or less | By design; charge about 3.2 h (estimate)              | Datasheets                         |
| R14 | Environment                   | Operate from 0 to 40 °C; electronics IP54 or better; survive rain on outdoor stoops                                   | By design; unverified                                 | Enclosure review; later spray test |
| R15 | Stair and building protection | No steel contact with stairs or walls in normal use; non-marking wheels; skids over stair nosings                     | By design; sprocket clearance over nosings unverified | Model check                        |

## ■ Requirements not met or at risk

- **R11 height is not met.** The handle must stand about 1,230 mm above the floor when the truck is tilted 30 degrees for climbing, which puts the upright height at about 1,460 mm. A telescoping or folding handle would fix this but adds cost and mass (see SCM-PRC-001, key design choices).
- **R2 is at risk.** A tri-star cluster is sized for one stair geometry. The proposed cluster (234 mm wheel spacing) targets residential stairs; steeper or shallower stairs may make it slip or bump.
- **R3, R5 and R12 have thin margins** of a few percent on the estimates.
- R6 to R9 depend on control and brake behavior that exists only as a concept.

## ■ Assumptions

- Payload 60 kg; truck about 24.3 kg (mass roll-up in SCM-PRC-001); total 84 kg.
- The clusters carry the whole weight while lifting (conservative; the operator's grip carries some in practice).
- Load center of mass about 500 mm above the cluster shaft; frame tilted about 30 degrees back while climbing.
- Drive efficiencies: motor driver 95 %, brushed motor 80 %, self-locking worm 45 %, chain and bearings 95 %.
- Rolling resistance coefficient about 0.03 for solid rubber wheels on a smooth floor.
- Stair geometry from the US International Residential Code (196 mm maximum riser, 254 mm minimum tread); national and local codes vary.